

BEFORE SUBMITTING ANY QUESTION: re-read it and check **every** asked quantity has an answer · post-measurement states are **normalised** · notation as demanded (ket form, no $|+ \rangle|-\rangle$, no $\otimes$). *Three of your last four losses were unfinished answers, not wrong maths.*

1 · Basis order foundation

Index = binary. 2 qubits: $|00\rangle |01\rangle |10\rangle |11\rangle = e_0 e_1 e_2 e_3$. n qubits $\rightarrow 2^n$ entries. **Top wire = leftmost = most significant bit.**

2 · Valid state & norm 5 + 3 mk

Valid $\iff \sum |a_i|^2 = 1$. $|a+bi|^2 = a^2 + b^2$ (not $(a+bi)^2$). $|i/\sqrt{2}|^2 = \frac{1}{2}$.
Missing entry: $? = \sqrt{1 - \sum |\text{others}|^2}$ — any complex number of that modulus scores 100%.

Norm $\|v\| = \sqrt{\sum |v_i|^2}$. $\|(1,1)\| = \sqrt{2}$ · $\| |0\rangle + |1\rangle \| = \sqrt{2}$ (unnormalised!) · $\| |0\rangle \| = 1$ · $\|0\| = 0$ (the scalar).
TRAP $|0\rangle + |1\rangle$ is **not** $|+\rangle$ — there is no $1/\sqrt{2}$ unless it is written.
TRAP $\|0\|$ vs $\| |0\rangle \|$ are different questions. Read the bars.

3 · Tensor product $A \otimes B$ 5 mk

Every entry of **A** scales a **whole copy of B**, in order.
 $(a_0, a_1) \otimes (b_0, b_1, b_2, b_3) = (a_0 b_0, a_0 b_1, a_0 b_2, a_0 b_3, a_1 b_0, a_1 b_1, a_1 b_2, a_1 b_3)$ — $2 \cdot 4 = 8$ entries.
Matrices: $(n \cdot p) \times (m \cdot q)$; each a_{ij} scales a full copy of B.

TRAP $A \otimes B \neq B \otimes A$. Order reversal is the classic loss here.
TRAP Write **every** entry, including the zeros.

4 · Circuit tracing 8 mk

Full vector in fixed basis order $\rightarrow$ apply gates **strictly left to right** $\rightarrow$ write the state after **each** gate $\rightarrow$ stop **exactly** at the named marker. $\psi_1/\psi_2/\psi_3$ are graded separately.

gate	action
X	$[[0,1],[1,0]]$ — bit flip
Z	$[[1,0],[0,-1]]$ — phase flip
Y	$[[0,-i],[i,0]]$
H	$(1/\sqrt{2})[[1,1],[1,-1]]$
S / T	$\text{diag}(1, i)$ / $\text{diag}(1, e^{i\pi/4})$
CNOT(c $\rightarrow$ t)	flip t iff c = 1
Toffoli(a,b $\rightarrow$ t)	flip t iff a = b = 1

$H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ · $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ · **H|+⟩ = |0⟩ · X|+⟩ = |+⟩ · Z|+⟩ = |−⟩**
EX $|00\rangle \xrightarrow{H(\text{top})} (|00\rangle + |10\rangle)/\sqrt{2} \xrightarrow{\text{CNOT}(\text{top} \rightarrow \text{bot})} (|00\rangle + |11\rangle)/\sqrt{2} \xrightarrow{X(\text{top})} (|10\rangle + |01\rangle)/\sqrt{2}$

TRAP Keep the $2^{-n/2}$. $H^{\otimes n}$ on $|0^n\rangle$ makes 2^n terms.

5 · Measurement — FULL 4 mk

$P(b) = |a_b|^2$. Post-state = $|b\rangle$ (drop the phase; already normalised).
EX $\psi = (-i/2, 1/\sqrt{2}, 0, i/2) \rightarrow P(00) = |-i/2|^2 = \frac{1}{4}$, post = **(1,0,0,0)**

6 · Measurement — PARTIAL 4 mk

1. Zero out every component not in the outcome set **A** $\rightarrow$ unnormalised **v**.
2. $P(A) = \|v\|^2 = \sum_{i \in A} |a_i|^2$.
3. Post-state = **v** / $\|v\|$.
EX same ψ , $A_2 = \{01, 10\} \rightarrow P = |1/\sqrt{2}|^2 + 0 = \frac{1}{2}$, post = **(0,1,0,0)**

TRAP ALWAYS write the post-measurement state — omitting it is your single most expensive habit.

TRAP If both the non-normalised and the normalised state are asked, give **both**.

Two measurements: "without knowing the 1st" = **marginalise** (sum over both first outcomes). "Knowing the 1st" = **collapse first, then square**, and renormalise.

7 · Entanglement test (2 qubits) in T/F

(a_0, a_1, a_2, a_3) is a **product state** $\iff a_0 a_3 - a_1 a_2 = 0$. Non-zero $\Rightarrow$ entangled.

EX $\frac{1}{2}(1, -1, 1, -1): (\frac{1}{2})(-\frac{1}{2}) - (-\frac{1}{2})(\frac{1}{2}) = 0 \Rightarrow$ **NOT entangled** (it is $|+\rangle \otimes |-\rangle$)

8 · Hamiltonian $\rightarrow$ gate 8 mk

Time-independent Schrödinger $\hat{H}\psi = E\psi \Rightarrow E$ are the **eigenvalues**, ordered $E_0 \leq E_1$.
 $\hat{H} = X = [[0,1],[1,0]] \Rightarrow E_0 = -1, E_1 = +1$ (eigenvectors $|-\rangle, |+\rangle$).
 $U(t) = e^{-i\hat{H}t}$. If $\hat{H}^2 = I$: **$e^{-i\hat{H}t} = \cos(t) \cdot I - i \sin(t) \cdot \hat{H}$**
 $\Rightarrow$ acts as X (up to global phase) when $\sin t = \pm 1 \Rightarrow t = \pi/2$.

General: diagonalise $\hat{H} = PDP^\dagger$, then $U = P \cdot e^{-iD^\dagger t} \cdot P^\dagger$ — exponentiate the **diagonal only**, never entrywise.

TRAP The minus sign in $e^{-i\hat{H}t}$ is load-bearing.
TRAP A global phase is **not** a different gate — factor it out before comparing.

9 · Oracle circuit $\rightarrow$ f 5 mk

Track the **aux** wire. CNOT(a $\rightarrow$ aux) XORs a; Toffoli(a,b $\rightarrow$ aux) XORs a·b. f = whatever aux holds **at the CNOT(aux $\rightarrow$ y)**. Everything after is **uncomputation — ignore it**.
EX CNOT($x_1 \rightarrow$ aux), Tof($x_2, x_3 \rightarrow$ aux), CNOT(aux $\rightarrow$ y), then the mirror image $\Rightarrow f = x_1 \oplus (x_2 \wedge x_3)$

TRAP Counting the uncomputation half doubles your terms.

10 · Bernstein-Vazirani 8 + 4 mk

$|0^n\rangle - H^{\otimes n} - U_f - H^{\otimes n} -$ measure **s** ; $|1\rangle - H - U_f$. Promise: $f(x) = x \cdot s$.
Read s off a truth table: $s_i = f(e_i)$ — evaluate f on 100..., 010..., 001....

EX $f(100)=1, f(010)=1, f(001)=0 \Rightarrow s = 110$

$f \equiv 0 \rightarrow s = 0^n$. $f \equiv 1 \rightarrow s = 0^n$ (global -1 only). $f = \neg(x \cdot s) \rightarrow$ still s .

- Which modifications still give the same s?**
- YES** — measure bottom wire **AFTER** U_f : it holds $|-\rangle$ and is already unentangled from the answer.
 - NO** — measure bottom wire **BEFORE** U_f : destroys $|-\rangle \Rightarrow$ no phase kickback.
 - YES** — replace the **1st** $H^{\otimes n}$ by DFT_{2^n} : both build the uniform superposition from $|0^n\rangle$.
 - NO** — replace the **2nd** $H^{\otimes n}$ by DFT_{2^n} : the 2nd layer must *undo* the 1st, and $DFT \neq H^{\otimes n}$ for $n > 1$.
- TRAP** An *ignored* measurement still collapses the state. "Ignored" $\neq$ "harmless".
TRAP The two H layers do **different jobs**; a swap safe on layer 1 is not safe on layer 2.

11 · DFT 3 mk

$\text{DFT}_N|0\rangle$ = uniform superposition (= $H^{\otimes n}|0^n\rangle$), and **not** entangled).

Input comb: a non-zero every r -th slot in $\mathbb{C}^N \Rightarrow$ the DFT is a comb of period $t = N / r$.

EX non-zero every 4th entry, $N = 32 \Rightarrow t = 8$

TRAP The periods are **inverse** — a tighter input comb gives a wider output comb.

12 · Shor 8 mk

$f(x) = a^x \bmod N$. $r = \text{ord}(a) \bmod N = \text{smallest } r > 0 \text{ with } a^r \equiv 1 \pmod N$.

EX $5^1=5, 5^2=25, 5^3=125\equiv 1 \pmod{124} \Rightarrow \text{ord}(5) \bmod 124 = 3$. $124 \equiv 4 \pmod 5, 4^2=16\equiv 1 \Rightarrow \text{ord} = 2$.

Upper-register outcome $c \approx$ a multiple of $2^n / r$. ($n = 100, r = 3 \Rightarrow 2^{100}/3$.)

Then continued fractions on $c/2^n$ recover r . **Factoring:** if r is even and $a^{r/2} \not\equiv -1 \pmod N$, take $\text{gcd}(a^{r/2} \mp 1, N)$ and report the non-trivial one; otherwise the attempt **fails** — say so.

TRAP Check the odd- r and the -1 failure conditions **before** computing any gcd.

13 · Grover in T/F

$t_{\text{opt}} \approx (\pi/4) \cdot \sqrt{(N/M)}$, $N = 2^n$, M = number of solutions. Queries $O(\sqrt{N}) = 2^{n/2}$.

- More iterations does **NOT** necessarily help — success probability **oscillates**. **(F)**
- Even at the optimal count, success probability is **not exactly 1** in general. **(F)**
- A t tuned for **one** solution is wrong for **two** — it overshoots. **(F)**

TRAP $\sqrt$ of the **search space**, not of n . An $O(n^2)$ oracle multiplies total *time*, not the *query count*.

14 · Stabilisers 6 + 4 mk

Set $\rightarrow$ state: each generator is a $+1$ -eigenvalue constraint. $Z: |0\rangle = +1, |1\rangle = -1$. $X: |+\rangle = +1, |-\rangle = -1$.

EX $\{ZI, -IZ\}$: $ZI \Rightarrow$ qubit 1 = $|0\rangle$; $-IZ \Rightarrow$ qubit 2 = $|1\rangle \Rightarrow |01\rangle$

Unitary $\rightarrow$ new set: conjugate every generator, **$S \rightarrow USU^\dagger$** (not US). Apply U only on the tensor factors it touches. Carry **every sign**.

U	conjugation
H	$X \leftrightarrow Z, Y \rightarrow -Y$
S	$X \rightarrow Y, Y \rightarrow -X, Z \rightarrow Z$
CNOT($c \rightarrow t$)	$XI \rightarrow XX, IZ \rightarrow ZZ, IX \rightarrow IX, ZI \rightarrow ZI$

EX $\{XX, -YZ\}$ under $H \otimes I$: $XX \rightarrow ZX$; $-YZ \rightarrow -(-Y)Z = +YZ \Rightarrow \{ZX, YZ\}$

TRAP Signs are load-bearing: $-Y$ stays negative unless the conjugation flips it.

15 · Error correction 3 mk

Shor code = 9 qubits, corrects **any single-qubit error**.

Error discretisation: every 1-qubit operation is a linear combination of I, X, Y, Z . So correcting X, Y and $Z \Rightarrow$ correcting **all** single-qubit errors. Hence "corrects X, Y, Z but not H -errors" is **FALSE**.

Transversal gate acts qubit-wise $\Rightarrow$ maps product states to product states $\Rightarrow$ if ψ is non-entangled then $G\psi$ is non-entangled. **TRUE**. (Eastin-Knill: no code has a *universal* transversal gate set.)

TRAP $Z^{\otimes 9}$ on $|0\rangle$ is a **logical operation**, not a detectable error. Check all three blocks.

16 · Gate sets / universality 4 mk

SWAP = 3 CNOTs $\Rightarrow$ you need an **entangling 2-qubit gate**. No entangler $\Rightarrow$ SWAP is impossible.

- $\checkmark$ Clifford (CNOT, H, S) $\cdot \checkmark$ Clifford + T $\cdot \checkmark$ {Toffoli, H} (universal; Toffoli with a $|1\rangle$ aux = CNOT)
 - $\times$ $\{X, Y, Z, S, T, H\}$ — all single-qubit, no entangler $\Rightarrow$ never.
- Universality:** finite 1-qubit sets like $\{X, Y, Z\}$ are not dense $\Rightarrow$ not universal even on one qubit. $R_X, R_Y, R_Z \Rightarrow$ dense on 1 qubit. $\{CNOT, H, T\} \Rightarrow$ fully universal.

17 · Uniform superposition 4 mk

$\sum_x 2^{-n/2}|x\rangle$ needs **every** basis state at **equal amplitude AND equal sign**.

- $\checkmark$ $H^{\otimes n}$ on $|0^n\rangle \cdot \checkmark$ DFT_{2^n} on $|0^n\rangle$
- $\times$ $X^{\otimes n}$ on $|0^n\rangle$ — X only permutes basis states, giving $|1^n\rangle$
- $\times$ H on one wire of $|+\rangle^{\otimes n} = H|+\rangle = |0\rangle$, kills the superposition on that wire
- $\times$ Z on one wire — $|+\rangle \rightarrow |-\rangle$, the sign flip breaks uniformity
- note** $X|+\rangle = |+\rangle$, so an X on an already-uniform wire is harmless

TRAP $|+\rangle$ on every wire is **already** uniform — the question is whether the extra gate breaks it.

18 · Fact bank for True/False 7 + 7 mk

- A valid quantum operation $\iff$ **unitary** ($U^\dagger U = I$, orthonormal columns). The all- $1/\sqrt{2}$ matrix has two equal columns $\Rightarrow$ **FALSE**.
- Two identical consecutive measurements give the same outcome. **TRUE** (the state already collapsed).
- ... with a **Z** in between: still the same. **TRUE** (Z is diagonal in the computational basis — phase only).
- ... with an **X** in between: **FALSE** (X flips the outcome).
- A finite continued fraction is always rational. **TRUE**.
- $\text{DFT}_{2^n}|0^n\rangle$ is entangled? **FALSE** — it is $|+\rangle^{\otimes n}$, a product state.
- A global phase never changes a measurement outcome.
- Measurement is irreversible; an ignored outcome still collapses the state.

19 · Pacing 90 min · 100 mk · 19 Q

≈ 4.7 min per question · 54 s per mark. The real paper is 19 substantial multi-part questions, not a rapid-fire e-test — you have more time per item than the practice e-tests trained you for. Use it to **finish every part**.

Order of attack: ① the T/F blocks and the short recalls (norms, DFT period, validity, gate sets, uniform superposition) — ~ 25 marks in ~ 15 min. ② the mechanical computes (tensor, circuit trace, measurement, stabilisers, oracle $\rightarrow$ f). ③ Hamiltonian and Shor last.

Partial credit is real — sub-parts are graded at 25/33/50% each. A half-finished derivation still scores; a blank does not. Never leave a box empty.